NLP Assignment Report

Part 3: Tokenization Gap

BERT Sentiment Analysis on IMDb Dataset

Roll Numbers Ending with 2 and 7 | Model: bert-base-uncased | Dataset: IMDb
Training Samples: 2,000 | Test Samples: 500 | Max Sequence Length: 128

1. Introduction

This report presents an experimental investigation into the Tokenization Gap in BERT-based models for sentiment analysis. The central research question is: does the choice of tokenization strategy meaningfully impact downstream classification performance when the underlying model architecture remains fixed?

WordPiece tokenization — the default strategy in BERT — is static: its vocabulary is fixed at pre-training time and may not be optimal for every downstream corpus. This study evaluates four tokenization strategies against the IMDb movie review dataset using bert-base-uncased as the shared backbone.

2. Objective

The experiment aims to:

- **Replace WordPiece** with alternative tokenization schemes: Byte Pair Encoding (BPE) and Character-Level tokenization.
- **Evaluate the performance impact** of each scheme on accuracy, F1 score, training time, and model size.
- **Implement extensions:** Hybrid tokenization (word + character fallback) and Dynamic Token Merging.
- **Compare all strategies** to identify which approach best balances accuracy and efficiency.

3. Experimental Setup

3.1 Hardware & Software

- **Hardware:** Google Colab with NVIDIA T4 GPU, mixed-precision (fp16) training.
- **Model:** bert-base-uncased (~418 MB, 110M parameters).
- **Framework:** HuggingFace Transformers, Tokenizers, PyTorch, scikit-learn.
- **Dataset:** IMDb (HuggingFace datasets), shuffled with seed=42 for balanced class distribution.

3.2 Hyperparameters

Parameter	Value
Training samples	2,000 (shuffled)
Test samples	500 (shuffled)
Max sequence length (MAX_LEN)	128 tokens
Training epochs	2
Train batch size	16
Eval batch size	32
Precision	fp16 (mixed precision on GPU)
Evaluation strategy	Per epoch

4. Tokenization Strategies

4.1 WordPiece (Baseline)

WordPiece is BERT's native tokenization algorithm. It segments words into subword units using a greedy longest-match approach on a fixed pre-trained vocabulary (~30,000 tokens). It is purpose-built for BERT's architecture and serves as the performance ceiling baseline in this study.

4.2 Byte Pair Encoding (BPE)

BPE learns a vocabulary of subword merge rules from the training corpus itself. Starting from individual characters, it iteratively merges the most frequent adjacent pair until the target vocabulary size (10,000 in this experiment) is reached. BPE is data-driven and better reflects corpus-specific morphology compared to the fixed WordPiece vocabulary.

4.3 Character-Level Tokenization

Each character (including spaces and punctuation) is treated as an individual token. This approach has a minimal vocabulary (~90 symbols) but generates extremely long sequences — averaging 1,479 tokens per document — far exceeding the MAX_LEN=128 limit. As a result, most of each review is truncated, causing a severe information loss that directly impacts classification accuracy.

4.4 Hybrid Tokenization (Extension)

The hybrid approach builds a word-level vocabulary from the training corpus. Known words are passed through as single tokens, while unknown (out-of-vocabulary) words are split to their constituent characters. This balances vocabulary coverage with sequence compactness, averaging only 67.7 tokens per document — the most compact representation among all four strategies.

4.5 Dynamic Token Merging (Extension — Demo)

Applied as a post-processing step to character-level sequences, dynamic merging iteratively fuses adjacent short tokens (length ≤ 2) into combined units. This reduces the extreme sequence

lengths produced by character-level tokenization, partially recovering information that would otherwise be lost to truncation.

5. Results

5.1 Main Performance Results

All four tokenizers were evaluated on the IMDb test set after 2 epochs of fine-tuning on bert-base-uncased.

Tokenizer	Accuracy	F1 Score	Time (s)	Size (MB)
WordPiece (Baseline)	0.8520	0.8518	45.5	418
BPE	**0.8620**	**0.8619**	44.7	418
Character-Level	0.5080	0.3423	46.7	418
Hybrid (Word + Char)	0.8260	0.8258	45.3	418

BPE (highlighted in green) achieved the best accuracy and F1 score. All tokenizers share the same 418 MB model backbone — differences are purely due to tokenization.

5.2 Tokenization Statistics (100 samples)

The table below shows the average token count per document and the effective vocabulary size seen in 100 training samples.

Tokenizer	Avg Tokens / Doc	Vocab Size
WordPiece	347.6	5,320
BPE	351.0	5,129
Character-Level	1,479.3	90
Hybrid	67.7	1,895

Character-Level produces sequences 4× longer than WordPiece or BPE, making the MAX_LEN=128 truncation especially damaging. Hybrid achieves the shortest average sequence (67.7 tokens) by keeping known words intact.

5.3 Dynamic Token Merging — Demo

Input sentence: "The movie was absolutely brilliant!"

Tokenizer	Tokens

WordPiece	['the', 'movie', 'was', 'absolutely', 'brilliant', '!']
BPE	['The', 'movie', 'was', 'absolutely', 'brilliant', '!']
Char-Level	['T','h','e',' ','m','o','v','i','e',' ','w','a','s', ...]
Dyn. Merge	['Th','e ','mo','vi','e ','wa','s ','ab','so','lu','te','ly',' b','ri','ll','ia','nt','!']

Dynamic merging halves the character-level token count by fusing adjacent short tokens, reducing sequence length and mitigating truncation loss.

6. Analysis & Discussion

6.1 BPE Outperforms WordPiece

BPE achieved the highest accuracy (0.8620) and F1 (0.8619), marginally surpassing WordPiece (0.8520 / 0.8518). This suggests that data-driven subword merges trained on the IMDb corpus capture domain-specific morphology slightly better than BERT's general-purpose pre-trained vocabulary. The performance gap is small (~1%), indicating both approaches are well-matched for sentiment classification.

6.2 Character-Level Fails Due to Truncation

Character-Level tokenization produced a dramatically lower F1 score (0.3423), falling below random chance for a binary classification task. The root cause is not the character-level representation itself, but the extreme mismatch between average document length (1,479 tokens) and the MAX_LEN=128 cap. Only the first ~50 characters of each review are retained after truncation, stripping the model of nearly all sentiment signal.

6.3 Hybrid Balances Accuracy and Efficiency

Hybrid tokenization achieved 0.8260 accuracy (0.8258 F1) with the most compact sequences (67.7 avg tokens). While it underperforms BPE, it offers a practical middle ground: compact sequences reduce memory and compute requirements, while the word-level backbone preserves semantic context that character-level tokenization loses.

6.4 Training Time and Model Size

All four experiments completed within 44–47 seconds per training run on T4 GPU, and all share the identical bert-base-uncased backbone (418 MB). Training time differences are negligible. This confirms that the performance differences are entirely attributable to the tokenization strategy, not to architectural changes.

7. Conclusion

This experiment demonstrates that tokenization strategy has a significant and measurable impact on BERT fine-tuning performance for sentiment analysis, even when the model architecture, training procedure, and dataset are held constant.

- **Best accuracy/F1:** BPE (0.8620 / 0.8619) — data-driven subword merges adapted to the IMDb corpus.
- **Fastest training:** BPE (44.7s) — marginally faster than others.
- **Most compact sequences:** Hybrid (67.7 avg tokens) — best for resource-constrained settings.
- **Worst performance:** Character-Level (F1 = 0.3423) — sequence truncation eliminates most review content.

The key lesson is that MAX_LEN compatibility is critical: any tokenization scheme that regularly produces sequences longer than the model's maximum must be paired with a larger MAX_LEN or a compression strategy (such as Dynamic Token Merging) to avoid truncation-induced accuracy collapse. For production deployment on IMDb-style long-form text, BPE is the recommended tokenizer.

8. References

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